

AI-Assisted Root Cause Analysis

Engineer Needs & Computational Requirements

Context: Nuclear Power Plant Operations

Version 1.0 | Project Inception Document

1. Purpose & Scope

This document establishes a comprehensive, integrated foundation of engineer needs and computational capability considerations for the development of AI-, ML-, and statistically-enabled tools to support Root Cause Analysis (RCA) and troubleshooting in nuclear power plants.

It integrates:

- Fundamental systemic considerations for nuclear RCA (data, causality, uncertainty, interdependencies, human factors, safety context)
- Practical engineer-driven needs (data silos, regulatory defensibility, CAP integration, knowledge retrieval, structured methodology compliance)

This document is intended to serve as the **conceptual starting point for formal requirements drafting and software architecture design**.

The tool suite is explicitly **decision support**, not decision automation. Final technical and regulatory conclusions remain the responsibility of licensed engineers.

All development must remain consistent with industry guidance from the **Electric Power Research Institute (EPRI)**, the **Nuclear Energy Institute (NEI)**, and regulatory expectations of the **U.S. Nuclear Regulatory Commission (NRC)**

2. Operating Context: Why Nuclear RCA Is Uniquely Challenging

Root cause analysis in nuclear power plants differs fundamentally from generic industrial troubleshooting due to:

- Defense-in-depth safety philosophy
- Highly coupled, redundant system architectures
- Low failure frequency / high reliability regime

- Strict regulatory documentation and auditability requirements
- Multi-disciplinary engineering integration
- Long asset lifetimes and aging degradation modes
- Cultural emphasis on conservative decision-making

The computational system must therefore support:

- Technical rigor
- Traceability
- Uncertainty transparency
- Human primacy
- Conservative inference bias
- Long-term knowledge preservation

2.1 Engineer Needs Overview

The diagram below maps the six identified engineer need areas to the computational RCA tool and its derived capability requirements. Colored nodes represent engineer needs; gray nodes represent specific analytical capabilities derived from those needs.

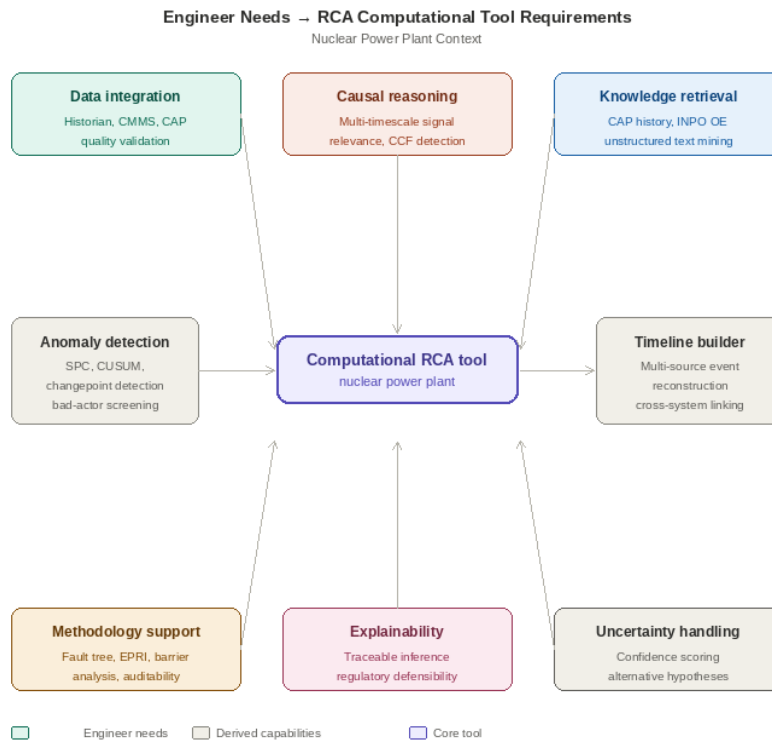


Figure 1. Engineer needs (colored) mapped to core RCA tool capabilities. Arrows indicate the need drives the capability requirement.

3. Critical Engineer Need Areas

The following sections describe each need area in detail, including the specific pain points that engineers encounter and the computational capabilities required to address them.

3.1 Data Access, Quality, and Integration

Problem Landscape

Plant-relevant information resides across fragmented systems:

- Process historians (high-frequency time series)
- Maintenance management systems
- Surveillance and testing databases
- Corrective Action Program (CAP) databases
- Operating experience (OE) repositories
- Protection system event logs
- Operator narrative logs
- Paper and legacy records

Engineers spend disproportionate time assembling and cleaning data rather than performing causal reasoning.

Key challenges:

- Timestamp misalignment
- Heterogeneous sampling rates
- Sensor drift, frozen values, rail conditions
- Outage gaps
- Cross-system entity disconnection
- No unified ontology linking equipment, work orders, and CAP records

Required Foundational Capabilities

- Multi-source ingestion framework with provenance tracking
- Unified timebase alignment across heterogeneous systems
- Automated data quality scoring and bad-actor detection
- Signal health monitoring and sensor reliability indicators
- Cross-system entity resolution (tag → equipment → work order → CAP record)
- Version-controlled data snapshots for RCA defensibility

This is the **non-negotiable Phase 1 foundation**. No analytical reliability exists without it.

3.2 System Topology, Functional Modeling & Defense-in-Depth Mapping

Problem Landscape

Nuclear plants are:

- Redundant (A/B trains)
- Cross-tied
- Interlocked through permissives and trips
- Supported by shared auxiliary systems
- Structured around safety functions

Symptoms frequently appear far from initiating causes.

Engineers must reason across:

- Physical configuration
- Functional relationships
- Protection logic
- Safety barriers

Required Capabilities

- Graph-based system topology model (equipment, systems, safety functions)
- Representation of redundancy and independence assumptions
- Protection logic modeling (permissives, trips, interlocks)
- Automatic mapping of affected safety functions
- Defense-in-depth barrier identification (which barriers failed, which succeeded)
- Visualization of degraded defense layers

This enables movement from correlation analysis to causal modeling grounded in plant architecture.

3.3 Signal-to-Noise & Event Identification

Problem Landscape

A single unit may contain 20,000–40,000 instrument channels.

Challenges include:

- Alarm flooding
- First-out signal identification
- Consequential alarm cascades
- Distinguishing symptoms from causes
- Rare-event detection
- Multi-timescale causality (milliseconds to months)

Engineers face:

- Proximate cause trap
- Confirmation bias
- Blindness to common-cause failures (CCF)

Required Capabilities

- Alarm clustering and cascade suppression
- First-out event detection logic
- Statistical Process Control (SPC)
- CUSUM for slow drift detection
- Changepoint detection
- Cross-channel correlation analysis
- Granger causality exploration (with caution)
- CCF screening against independence assumptions
- Multi-timescale pattern detection

The system must reduce cognitive overload without hiding critical signals.

3.4 Temporal Reconstruction & Timeline Intelligence

Problem Landscape

Causality is temporal.

Engineers often manually reconstruct:

- Event chronology
- Protection actuations
- Operator actions
- Maintenance overlaps
- Repeated intermittent failures

This is time-intensive and error-prone.

Required Capabilities

- Automated multi-source event timeline construction
- Millisecond-resolution sequence reconstruction (where data allows)
- Slow degradation visualization across weeks/months
- Comparative timeline matching with historical events
- Highlighting of precursor signatures

Timeline reconstruction is one of the highest near-term ROI capabilities.

3.5 Rare Events & Low-Frequency Failure Environment

Problem Landscape

Nuclear plants operate in high reliability regimes.

Traditional ML struggles because:

- Labeled failure data is sparse
- Many failure modes are rare or novel

Required Capabilities

- Hybrid physics-informed + statistical modeling
- Bayesian inference frameworks
- Anomaly detection over classification reliance
- Transfer learning across fleet data
- Explicit uncertainty quantification
- Conservative bias in sparse-data contexts

Black-box inferences shall not be used for safety-significant conclusions without traceable evidence pathways.

3.6 Latent Degradation & Aging Detection

Problem Landscape

Gradual degradation precedes many failures:

- Fouling
- Calibration drift
- Valve stiction
- Aging cable insulation

- Procedural erosion

These signals are often weak and long-term.

Required Capabilities

- Long-horizon trend analysis
- Cross-cycle comparison
- Fleet-wide pattern mining
- Change detection under noisy conditions
- Early weak-signal amplification

This supports preventive correction rather than post-failure RCA only.

3.7 Institutional Knowledge Preservation & Retrieval

Problem Landscape

Critical knowledge exists in:

- CAP narratives
- Industry OE reports
- Historical RCAs
- Informal expert experience

Much of this is unstructured text and difficult to retrieve.

Required Capabilities

- NLP-based semantic search across CAP and OE records
- Similarity matching between current symptoms and historical cases
- Structured encoding of prior RCA findings
- Knowledge graph linking failure modes, contributors, corrective actions
- Retirement-resilient knowledge capture

The system becomes a long-term institutional memory.

3.8 Structured Methodology Compliance & Analytical Flexibility

Methodology frameworks shall be dynamically linked to the underlying system topology and safety function model.

Problem Landscape

RCA must follow structured methods:

- Fault trees
- Barrier analysis
- Change analysis
- Event and causal factor charting

However:

- Over-rigidity may obscure novel failure modes
- Regulatory defensibility is mandatory

Required Capabilities

- Computer-aided fault tree construction
- Barrier analysis templates integrated with system topology

- Change analysis workflow support
- Audit trail from raw data to final conclusion
- CAP-ready documentation export
- Evidence traceability for every inference

The tool must enhance rigor without constraining exploration.

3.9 Uncertainty Quantification & Hypothesis Management

Problem Landscape

Evidence is often incomplete or degraded.

Current RCA practice frequently forces binary conclusions.

Risks:

- Overstated certainty → under-scoped corrective actions
- Over-broad actions → wasted resources

Required Capabilities

- Probabilistic confidence scoring
- Explicit evidence-gap representation
- Alternative hypothesis ranking
- Divergence explanation between hypotheses
- Sensitivity analysis under data-quality assumptions
- Distinction between “evidence against” and “insufficient data”

Uncertainty must be visible, not hidden.

3.10 Cognitive Bias Mitigation

Problem Landscape

Engineers are subject to:

- Anchoring
- Confirmation bias
- Availability bias
- Authority bias

These can truncate causal exploration prematurely.

Required Capabilities

- Automatic generation of alternative hypotheses
- Structured “disconfirming evidence” surfacing
- Hypothesis comparison dashboards
- Blind analysis modes where appropriate

The system should act as a cognitive challenger.

3.11 Multi-Disciplinary Collaboration & Workflow Integration

Problem Landscape

RCA spans:

- Mechanical

- Electrical
- I&C
- Operations
- Human performance
- Chemistry

Current collaboration is fragmented.

Required Capabilities

- Shared hypothesis workspace
- Hypothesis ownership tracking
- Version control of causal branches
- Structured dissent/comment capability
- Engineer override logging with rationale capture

Collaboration strengthens institutional learning.

3.12 Risk Significance & PRA Integration (Strategic)

Problem Landscape

Beyond cause identification, plants must evaluate:

- Safety function impact
- Barrier effectiveness
- Risk significance

Corrective actions are often risk-informed.

Required Capabilities (Long-Term)

- Mapping of event impacts to safety functions
- Interface capability with PRA models
- Qualitative risk significance scoring
- Identification of barrier weaknesses
- Risk-informed corrective action prioritization

This is likely Phase 3+ but should inform architecture from the beginning.

4. Human–Machine Collaboration & Trust Calibration

Transversal requirement across all domains.

Design Principles

- Human primacy: engineer directs analysis
- Interrogability: “Why are you suggesting this?”
- Transparent uncertainty
- Clear boundaries of competence
- Conservative inference bias
- Graceful degradation under missing data

Required Interface Capabilities

- Drill-down evidence chains
- Plain-language explanation summaries

- Hypothesis workspace manipulation
- Explicit confidence labeling
- Override logging for continuous model improvement

Trust must be earned through transparency.

5. Analytical Operating Modes

The system must support distinct modes:

Rapid Triage Mode

- Immediate troubleshooting
- Alarm prioritization
- First-out identification
- Preliminary hypothesis surfacing

Deep Forensic RCA Mode

- Structured methodology integration
- Full uncertainty representation
- Documentation support
- Regulatory defensibility outputs

Architecture should allow progressive analytical refinement.

6. Model Governance & Lifecycle Management

Because this is nuclear: Baseline configuration management of plant model snapshots (design changes, ECNs, setpoint revisions).

Because this is nuclear:

- Algorithm version control required
- Configuration management
- Model validation & periodic re-validation
- Performance monitoring for drift
- Controlled update mechanisms
- Full audit logging of model changes

ML governance must be embedded from inception.

7. Cross-Cutting Non-Negotiables

Interpretability shall take precedence over marginal predictive performance gains from opaque models.

1. Regulatory defensibility
2. Explainability over black-box optimization
3. Cybersecurity and on-prem/private deployment capability
4. Auditability of all system actions
5. Conservative bias in sparse-data contexts
6. Graceful failure modes
7. No automated closure of RCA without human authorization

8. Foundational Capability Pillars (Architecture Guidance)

From these needs, six architectural pillars emerge:

1. Data Fusion & Quality Layer
2. Plant Topology & Functional Knowledge Graph
3. Temporal & Statistical Analytics Engine
4. Causal Reasoning & Hypothesis Management Layer
5. Institutional Knowledge & NLP Retrieval Engine
6. Explainable Interface & Collaboration Layer

All future requirements and code architecture decisions should map to one or more of these pillars.

This document is a living requirements input. It should be reviewed and updated at each major project milestone.